

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 21)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1

Describe how hydrothermal deposition produces exploitable concentrations of metal ores.

[4 marks]

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0 1 . 2

Explain how Proterozoic marine sediments produced the world's major iron ore deposits.

[2 marks]

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0 1 . 3

Explain how alluvial (physical sediment) processes separate and concentrate minerals such as gold and tin ore.

[2 marks]

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0 1 . 4

Give **two** examples of mineral deposits formed by biological sedimentation and state their origin.

[2 marks]

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0 2 . 1

Explain how affluence and the strategies of manufacturers increase the amount of solid waste produced.

[4 marks]

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0 2 . 2

Give **three** factors that affect the choice of disposal method for a solid waste.

[3 marks]

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0 2 . 3

Explain **one** environmental advantage and **one** disadvantage of composting garden and food waste rather than sending it to landfill.

[2 marks]

Advantage

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Disadvantage

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0 2 . 4 Explain why mixed municipal waste is difficult to recycle.

[1 mark]

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10

0 3 Figure 1 compares the energy flows in a conventional gas-fired power station and a combined heat and power (CHP) station, each supplied with 100 units of fuel energy.

Figure 1

	Conventional power station	CHP station
Electricity output / units	40	35
Useful heat supplied to buildings / units	0	45
Waste heat lost / units	60	20

0 3 . 1 Calculate the overall efficiency of each station.

[2 marks]

Answer

%

0 3 . 2 Explain why the electrical efficiency of a CHP station is often deliberately kept below 40%.

[2 marks]

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03 . 3

Explain why CHP stations must be located close to the users of the heat.

[1 mark]

03 . 4

Explain how each of the following features of an industrial heat exchanger increases its efficiency.

[3 marks]

Long narrow copper pipes

Counter-current flow

Reduced flow rate

03 . 5

Explain why a single large spherical tank loses less heat than several small cubic tanks holding the same total volume of hot fluid.

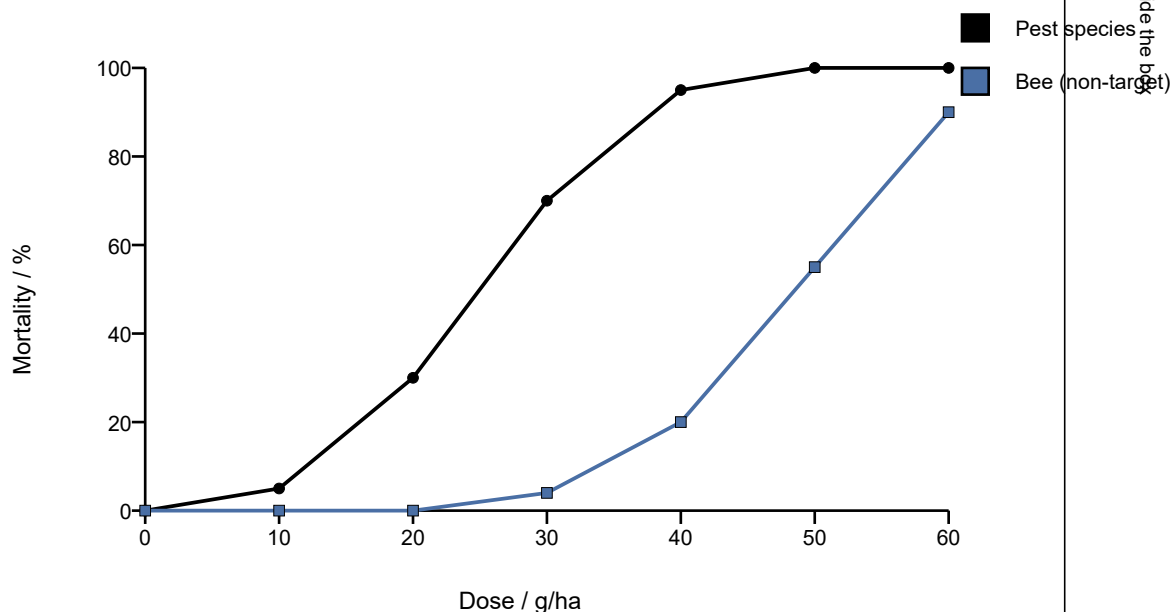
[2 marks]

10

0 4

Figure 1 shows how the percentage mortality of a pest species and of a non-target bee species change with the dose of an insecticide applied to a crop.

Figure 1



0 4

1

Use Figure 1 to estimate the LD_{50} of the insecticide for the pest species.

[1 mark]

Answer

g/ha

0 4

2

Define the term **MDNF** (maximum dose never fatal).

[1 mark]

0 4

3

Use Figure 1 to explain why it may not be possible to kill all of the pest species without harming bees.

[3 marks]

0 4 . 4

Suggest how the farmer could reduce the harm to bees without reducing the dose applied.

[2 marks]

0 4 . 5

Explain why a pesticide with high specificity is less likely to cause environmental damage.

[3 marks]

10

0 5

A student determined the water and organic matter content of a soil sample. Table 1 shows the results.

Table 1

Measurement	Mass / g
Empty crucible	25.0
Crucible + fresh soil	65.0
Crucible + soil after drying at 105 °C to constant mass	55.0
Crucible + soil after heating in a furnace at 550 °C	52.6

0 5 . 1

Calculate the percentage water content of the fresh soil.

[2 marks]

Answer

%

0 5 . 2

Calculate the percentage organic matter content of the dry soil.

[2 marks]

Answer

%

0 5 . 3

Explain why the soil is dried to constant mass before the organic matter is burnt off.

[2 marks]

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0 5 . 4

Explain how the organic matter content affects the fertility and erosion risk of this soil.

[3 marks]

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0 5 . 5

Suggest **one** reason why two students measuring the same field might get different organic matter results.

[1 mark]

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0 6

A student used 0.5 m x 0.5 m grid quadrats to survey ground flora in two meadows, one grazed and one ungrazed. Table 1 shows the results for the grazed meadow (10 quadrats).

Table 1

Species	Number of individuals
Ribwort plantain	48
Perennial ryegrass	120
White clover	36
Daisy	24
Creeping buttercup	12

0 6 . 1

Explain why a 0.5 m x 0.5 m quadrat was chosen rather than a 10 cm x 10 cm or 100 m x 100 m quadrat.

[2 marks]

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0 6 . 2

Calculate Simpson's Index of Biodiversity (D) for the grazed meadow using $D = N(N-1) / \sum n(n-1)$. Show your working.

[3 marks]

Answer

06 . 3

The ungrazed meadow had $D = 1.8$ with the same number of species. Explain what this shows.

[2 marks]

06 . 4

Explain why a point quadrat would be better than a frame quadrat in tall vegetation.

[2 marks]

06 . 5

Give **one** limitation of using the DAFOR abundance scale.

[1 mark]

10

07

Table 1 shows features of oil tanker design and operation that reduce the risk of oil pollution.

07 . 1

Complete Table 1 by explaining how each feature reduces oil pollution.

[5 marks]

Table 1

Feature	How it reduces oil pollution	
Double hull		
Inert gas system		
Separate oil and ballast tanks		
Twin engines / rudders		
Recirculation of oil during unloading (crude oil washing)		

07 . 2 Explain how each of the following reduces the risk of tanker accidents. **[2 marks]**

Offshore shipping routes

GPS / Automatic Identification System

07 . 3 Describe how a **bund wall** prevents oil pollution from a storage tank. **[2 marks]**

07 . 4 Explain the purpose of an oil interceptor on the drainage system of a car park. **[1 mark]**

10

0 8 . 1

Which of the following is a **secondary** pollutant? Tick one box.

[1 mark]

- A Sulfur dioxide from a coal-fired power station
- B Tropospheric ozone
- C Carbon monoxide from a car exhaust
- D Nitrogen monoxide from a diesel engine

☐
☐
☐
☐

0 8 . 2

Explain the difference between a primary pollutant and a secondary pollutant, using acid rain as an example.

[4 marks]

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0 9 . 1

Explain how a ground-source heat pump can deliver more heat energy to a building than the electrical energy used to run it.

[3 marks]

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0 9 . 2

Explain how Concentrating Solar Power (CSP) with thermal storage overcomes the intermittency of solar power.

[3 marks]

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09 . 3

Explain why CSP can only be used in areas with little cloud, whereas PV can be used anywhere.

[2 marks]

09 . 4

Describe **two** environmental impacts of solar PV.

[2 marks]

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2

10

10 . 1

Evaluate the success of the methods and strategies used to reduce global climate change.

[9 marks]

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10 . **2** State what is meant by a **carbon footprint**.

[1 mark]

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Write an essay on one of the following topics.

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Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment.

[25 marks]

OR

1 1 . 2

Discuss the strategies that can be used to provide sustainable supplies of water for a growing population.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS